

VIEWS

SIMPLE VIEW

1. Simple View (Single Table)

```
MySQL 8.0 Command Line Cli
Database changed
mysql> CREATE TABLE Student (
  ->   StudentID INT PRIMARY KEY,
  ->   Name VARCHAR(50),
  ->   Department VARCHAR(30),
  ->   Age INT
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Student VALUES
  -> (101,'John','CSE',20),
  -> (102,'Alice','ECE',21),
  -> (103,'David','CSE',22);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Student_View;
ERROR 1146 (42S02): Table 'dbms.student_view' doesn't exist
mysql> CREATE VIEW Student_View AS
  -> SELECT StudentID, Name, Department
  -> FROM Student;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Student_View;
+-----+-----+-----+
| StudentID | Name | Department |
+-----+-----+-----+
| 101 | John | CSE |
| 102 | Alice | ECE |
| 103 | David | CSE |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> |
```

2. Simple View (Multiple Tables)

```
MySQL 8.0 Command Line Cli
mysql> CREATE TABLE Course (
  ->   CourseID INT PRIMARY KEY,
  ->   CourseName VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Student_Course (
  ->   StudentID INT,
  ->   CourseID INT,
  ->   PRIMARY KEY(StudentID, CourseID),
  ->   FOREIGN KEY(StudentID) REFERENCES Student(StudentID),
  ->   FOREIGN KEY(CourseID) REFERENCES Course(CourseID)
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO Course VALUES
  -> (1,'DBMS'),
  -> (2,'Java');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Student_Course VALUES
  -> (101,1),
  -> (102,2),
  -> (103,1);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Student_Course_View;
ERROR 1146 (42S02): Table 'dbms.student_course_view' doesn't exist
mysql> CREATE VIEW Student_Course_View AS
  -> SELECT s.StudentID,
  ->        s.Name,
  ->        c.CourseName
  -> FROM Student s
  -> JOIN Student_Course sc
  -> ON s.StudentID = sc.StudentID
  -> JOIN Course c
  -> ON sc.CourseID = c.CourseID;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Student_Course_View;
+-----+-----+-----+
| StudentID | Name | CourseName |
+-----+-----+-----+
| 101 | John | DBMS |
| 103 | David | DBMS |
| 102 | Alice | Java |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> |
```

COMPLEX VIEW

3. Complex View (Single Table)

```
mysql> CREATE TABLE Employee (  
->     EmpID INT PRIMARY KEY,  
->     Name VARCHAR(50),  
->     Department VARCHAR(30),  
->     Salary DECIMAL(10,2)  
-> );  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> INSERT INTO Employee VALUES  
-> (1,'Ram','IT',50000),  
-> (2,'Priya','IT',60000),  
-> (3,'Arun','HR',45000),  
-> (4,'Meena','HR',55000);  
Query OK, 4 rows affected (0.01 sec)  
Records: 4 Duplicates: 0 Warnings: 0  
  
mysql> CREATE VIEW Department_AvgSalary AS  
-> SELECT Department,  
->     AVG(Salary) AS AverageSalary  
-> FROM Employee  
-> GROUP BY Department;  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> SELECT * FROM Department_AvgSalary;  
+-----+-----+  
| Department | AverageSalary |  
+-----+-----+  
| IT        | 55000.000000 |  
| HR        | 50000.000000 |  
+-----+-----+  
2 rows in set (0.00 sec)
```

4. Complex View (Multiple Tables)

```
mysql> CREATE TABLE Orders (
  ->   OrderID INT PRIMARY KEY,
  ->   CustomerID INT,
  ->   Amount DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE Customers (
  ->   CustomerID INT PRIMARY KEY,
  ->   CustomerName VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Customers VALUES
  -> (1,'Ravi'),
  -> (2,'Anita');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Orders VALUES
  -> (101,1,500),
  -> (102,1,800),
  -> (103,2,600);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE VIEW Customer_TotalOrders AS
  -> SELECT c.CustomerID,
  ->        c.CustomerName,
  ->        COUNT(o.OrderID) AS TotalOrders,
  ->        SUM(o.Amount) AS TotalAmount
  -> FROM Customers c
  -> JOIN Orders o
  -> ON c.CustomerID = o.CustomerID
  -> GROUP BY c.CustomerID, c.CustomerName;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Customer_TotalOrders;
+-----+-----+-----+-----+
| CustomerID | CustomerName | TotalOrders | TotalAmount |
+-----+-----+-----+-----+
| 1 | Ravi | 2 | 1300.00 |
| 2 | Anita | 1 | 600.00 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> |
```